Random Access Memories

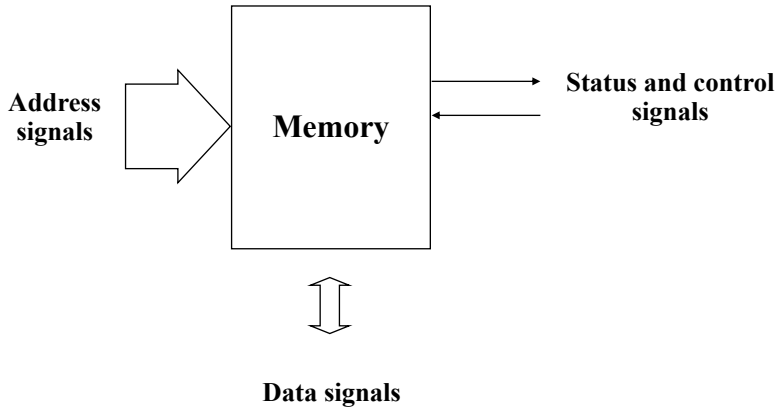
Matteo Sonza Reorda

Politecnico di Torino
Dip. di Automatica e Informatica



General characteristics

- **Each cell can be independently accessed**
- **The access time is constant and equal for all cells.**



Control signals

- They allow the memory to know
  - The requested operation type (read or write)
  - When the addresses are available on the Abus
  - When the data are available on the Dbus (for write operations).
- A mechanism should also be implemented to allow the external world to know
  - When the data produced by the memory are available (for read operations)
  - When it is possible to go ahead with a new memory access cycle.

3

Control sig

Synchronous solution: both the memory and the user share a clock signal and use it know when to go ahead

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- A mechanism should also be implemented to allow the external world to know
  - When the data produced by the memory are available (for read operations)
  - When it is possible to go ahead with a new memory access cycle.

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Control s

Asynchronous solution: the memory and the user do not share any clock signal. Hence, some additional control signals are required to allow them to know when to go ahead.

- They allow the memory to kn
  - The requested operation type (write)
  - When the addresses are avail n the Abus
  - When the data are availabl n the Dbus (for write operations).
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5

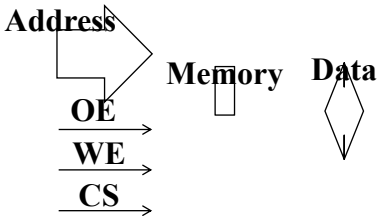
Control signals: examples

- Chip Select (CS):
  - It must be active to enable any operation on the memory
- Output enable (OE)
  - It must be active to enable the memory output signals (otherwise they hold the value Z)
- Write enable (WE)
  - It must be active to perform a write operation.

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Example

- To perform a write operation
  - Activate CS
  - Put the address on the Abus
  - Put the data on the Dbus
  - Activate WE
- To perform a read operation:
  - Activate CS
  - Put the address on the Abus
  - Activate OE
  - Read the data from the Dbus.



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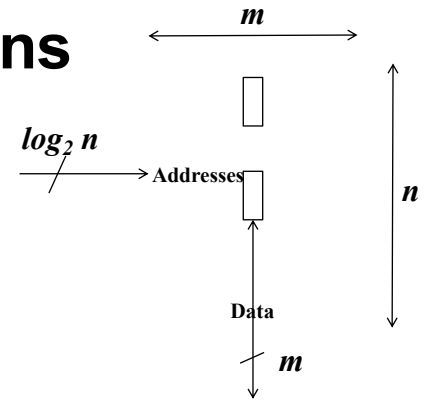
Status signals

Examples:

- Error
  - The memory tried to read a faulty word
- MFC
  - The memory completed the requested read / write operation.

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Dimensions



Memories are characterized by

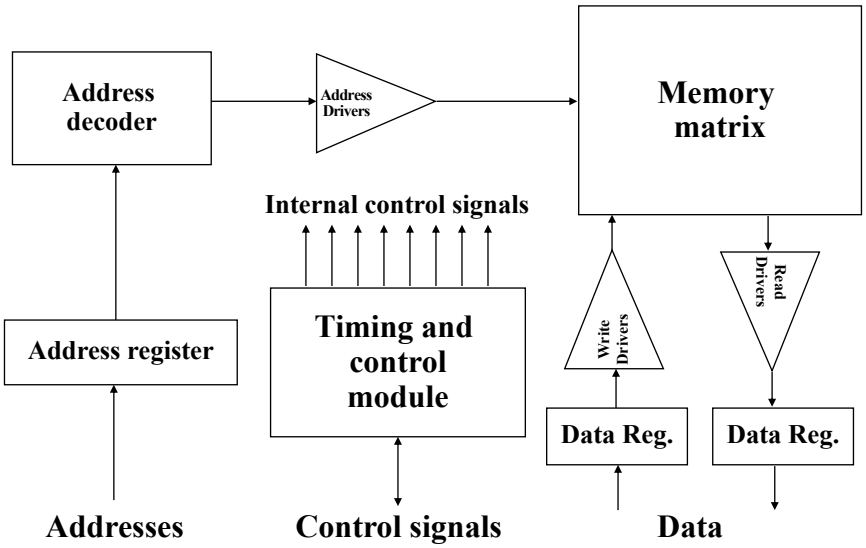
- Number of words (n)
- Number of bits per word (m).

The number of input/output signals is given by

- $log_2\ n$ address signals
- m data signals.

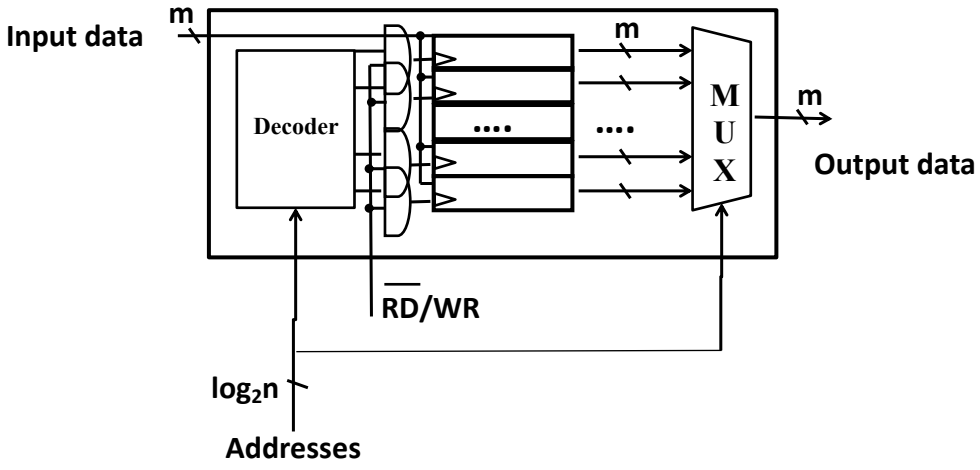
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Architecture



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General scheme



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Architecture

The cost of a RAM also depends on the complexity of the circuitry around the memory matrix.

This can be reduced through an appropriate memory cell organization.

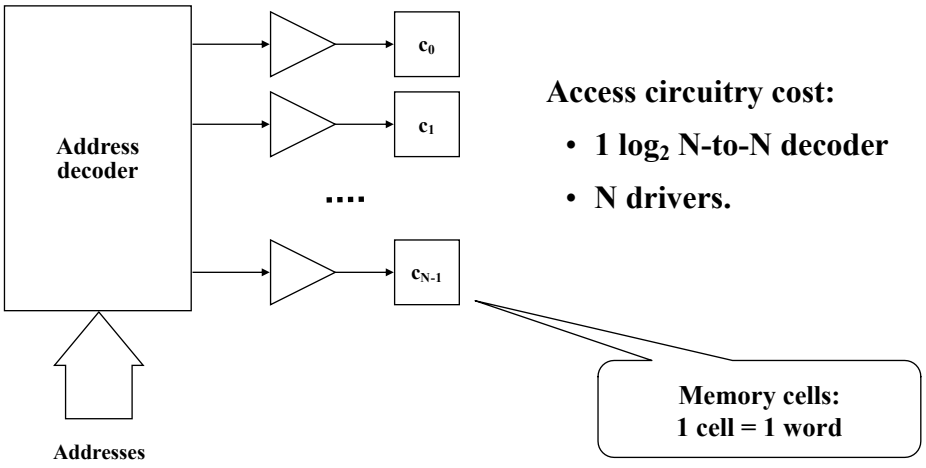
There are two main types:

- **vector organization**
- **two-dimensional organization.**

Regularity in the organization of memory cells also heavily influences the cost of layout.

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Vector organization



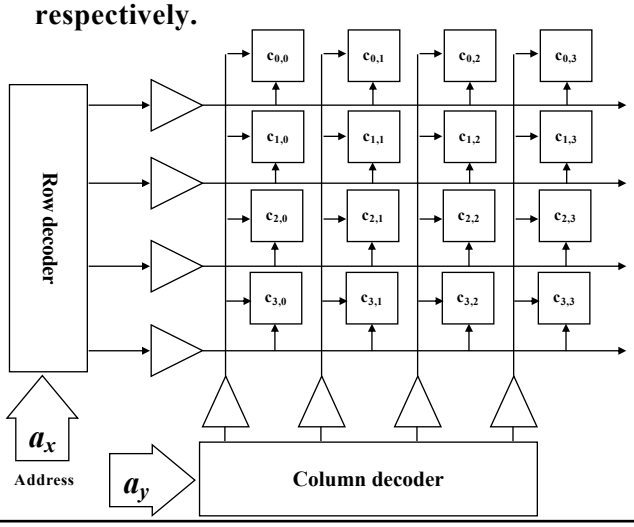
Access circuitry cost:

- **1 $\log_2$ N-to-N decoder**
- **N drivers.**

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Matrix organization

The address *a* is divided in 2 parts a_x and a_y, which select the row and column where the requested memory cell lies, respectively.



Access circuitry cost :

- **2 ($\log_2 \sqrt{N}$)-to-$\sqrt{N}$ decoders**
- **$2\sqrt{N}$ drivers.**

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Matrix organization: advantages

- **Lower cost of the HW**
- **More compact layout.**

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RAS and CAS signals

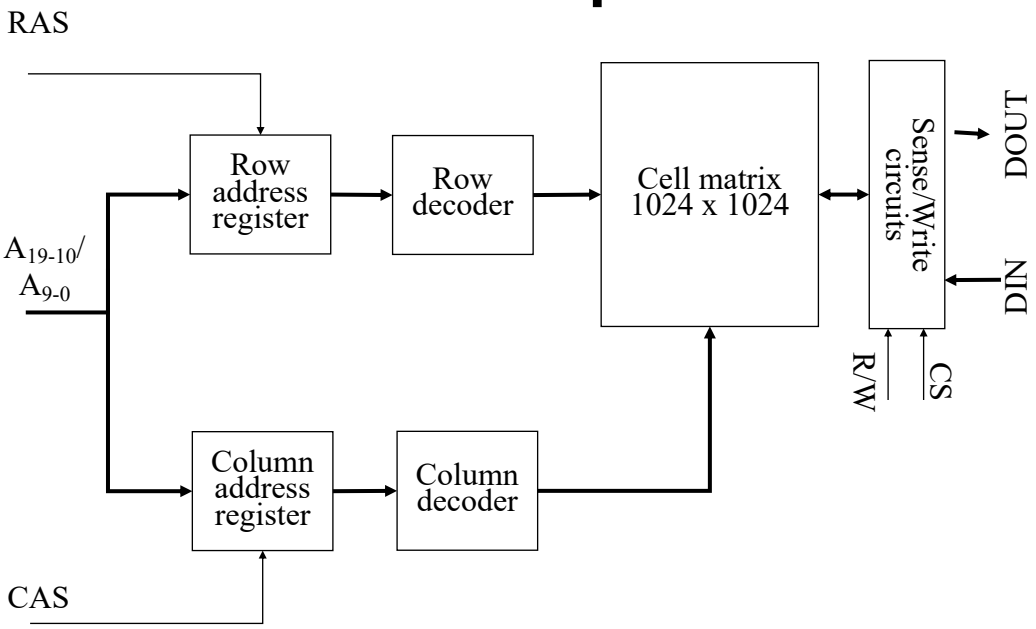
To reduce the number of input signals, RAMs using the matrix organization sometimes provide a different access protocol and a smaller number of address signals (approximately half).

In this case the address is provided in two distinct and subsequent steps, using the same address signals:

- **in a first phase, the address signals that go to the row decoder are provided, accompanied by the *Row Address Strobe* (RAS) signal**
- **in another phase, the address signals that go to the column decoder are provided, accompanied by the CAS (*Column Address Strobe*) signal.**

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Example



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Page Mode

If it is necessary to access consecutively a set of cells placed at successive addresses, and they reside in the same row of the matrix, you can enable the so-called *Page Mode*:

1. **the part of the address identifying the row is sent to the memory, accompanied by RAS**
2. **the part of the address identifying the column, accompanied by CAS, is sent to the memory**
3. **repeat from step 2.**

In this way you can reduce memory access times.

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Semiconductor memories

Memories are currently the most dense devices on ICs

By increasing the integration level it is possible to

- **increase capacity (with constant IC size)**
- **increase speed**
- **reduce consumption.**

Classification

- **ROM (*Read Only Memory*)**
- **PROM (*Programmable Read Only Memory*)**
- **EPROM (*Electrically Programmable Read Only Memory*)**
- **EEPROM (*Electrically Erasable Programmable Read Only Memory*)**
- **Flash**
- **RAM.**

ROM

Applications:

- **frequently used procedure libraries**
- **system programs**
- **function tables.**

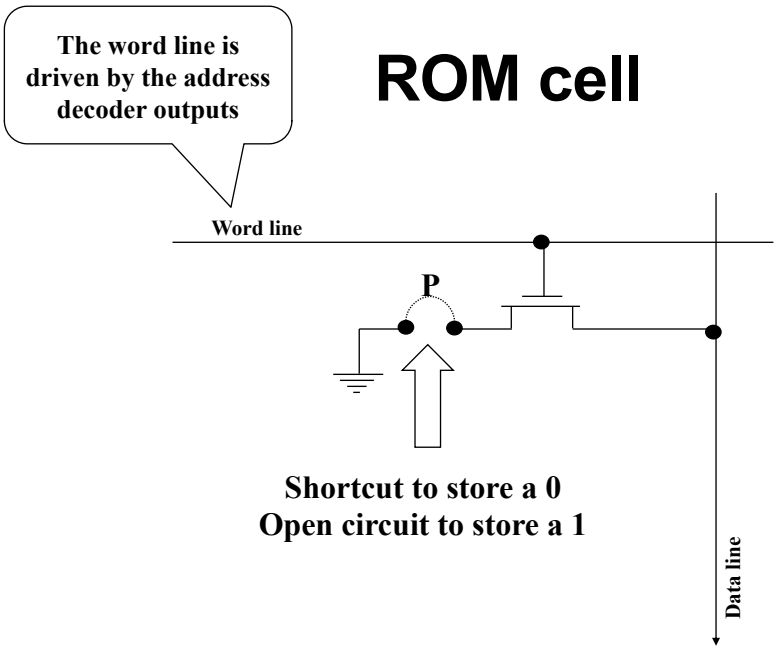
The definition of the content takes place before the silicon is produced, and the content is in no way modifiable later.

Cost is dominated by the fixed cost to set up the device's production line, and is almost independent of the number of chip produced.

The impact of any design error is dramatic.

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ROM cell



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PROM

Writing is done after the production process when the memory is already in the hands of the customer using special equipment called *programmers*.

Physically, PROMs are made by placing a diode in the point P of the previous scheme, which can be burned during programming, creating an open circuit.

Writing can only take place once.

The PROM is non-volatile.

They are preferable to ROMs for low volumes.

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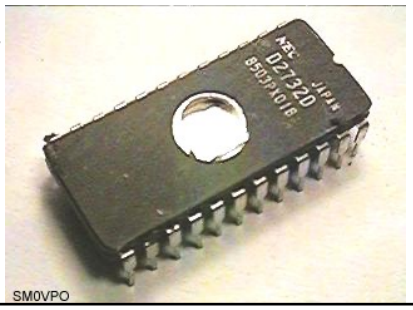
EPROM

They can be reprogrammed, after deletion by prolonged exposure (~ 20 min) to ultraviolet light.

Write can be done an indefinite number of times.

Write can also be made after mounting the device on the board.

They are more expensive than PROMs.



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EEPROM

EEPROMs (or E^2PROMs) can be reprogrammed byte by byte even after mounting on the board, but the operation takes longer than the reading (hundreds of μs).

Writing can be done through normal channels (buses) and signals.

They are more expensive and less dense than EPROMs.

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Flash memories

The cost is intermediate between EPROM and EEPROM.

They use only one transistor per bit, and are therefore relatively dense.

Read operations can be performed like in RAMs, while writing operations

- **are slower (at least one order of magnitude)**
- **require a previous deletion operation of the whole block.**

Flashes can also survive to a limited number of write cycles.

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Summary

Type	category	Erase	Write	Volatility
RAM	Read/Write	Electrical	Electrical	Volatile
ROM	Read-only	Impossible	At manufacturing	Non-volatile
PROM	Read-only	Impossible	Electrical	Non-volatile
EPROM	Read-mostly	UV light	Electrical	Non-volatile
EEPROM	Read-mostly	Electrical	Electrical	Non-volatile
Flash	Read-mostly	Electrical	Electrical	Non-volatile

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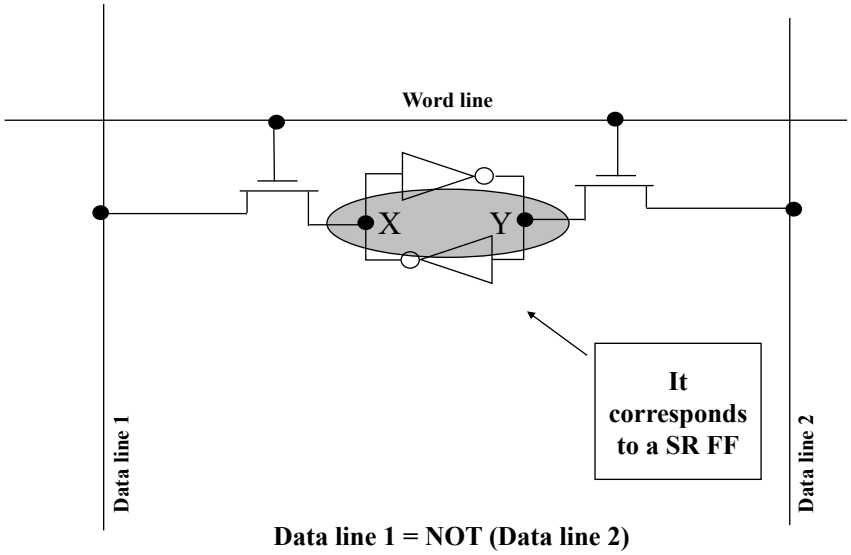
RAM

There are 2 types of RAMs:

- static RAMs (or SRAMs):
  - each cell corresponds to one flip flop
- dynamic RAMs (or DRAMs):
  - each cell corresponds to one capacitor and one transistor
  - the information is stored as a possibly charged capacitor
  - they require a periodic refresh of the information
  - the reading is destructive (*Destructive Read-Out*).

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Static RAM cell



Data line 1 = NOT (Data line 2)

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Behavior

To read or write a word, you must activate its word line.

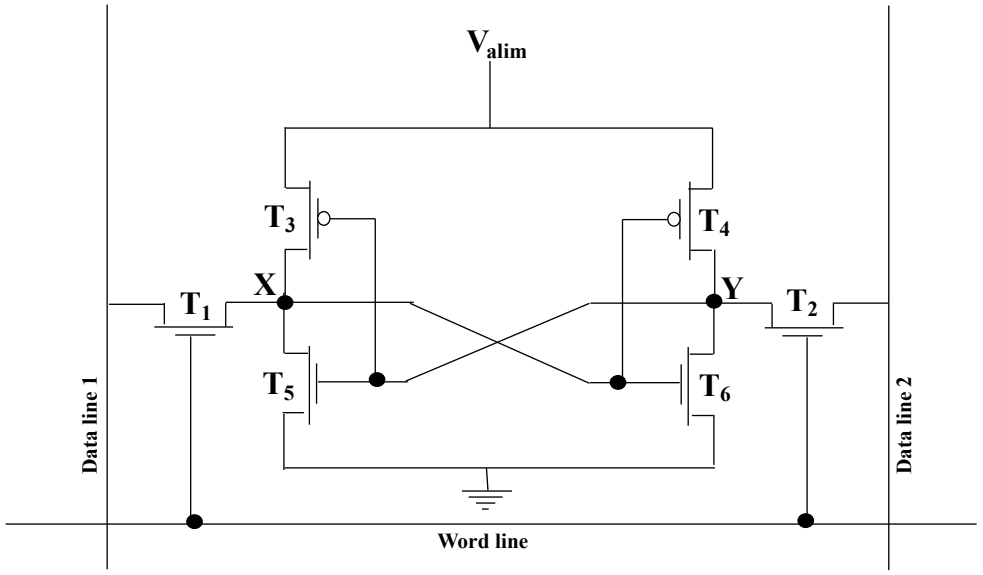
When the word line is not activated, its flip flop keeps its value.

When the word line is turned on, it is possible to

- read the (opposite) values forced by the flip flop on the two data lines (*reading*)
- write a new value, forcing two opposite values on the data lines (*writing*).

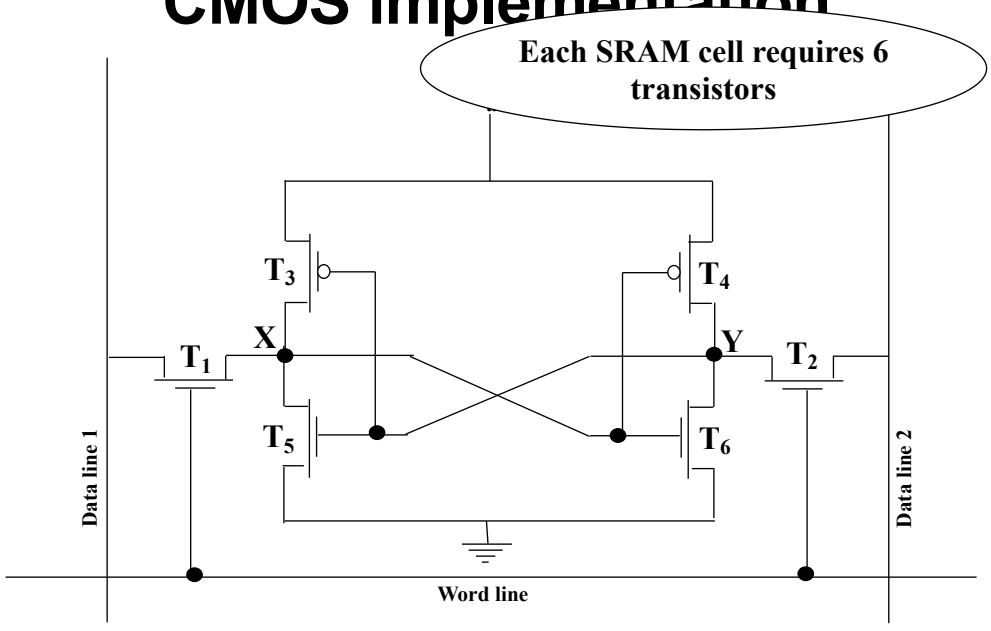
30

CMOS implementation



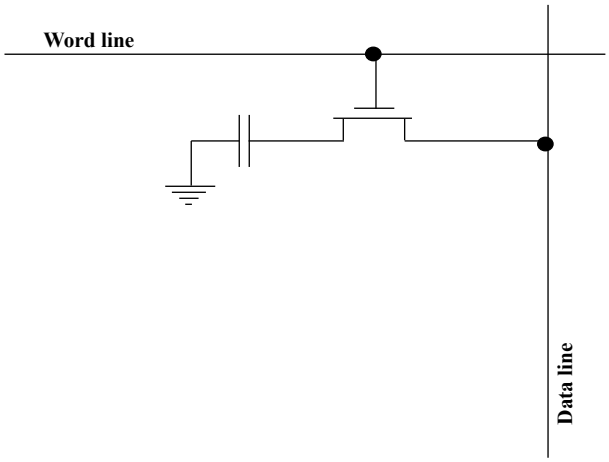
31

CMOS implementation



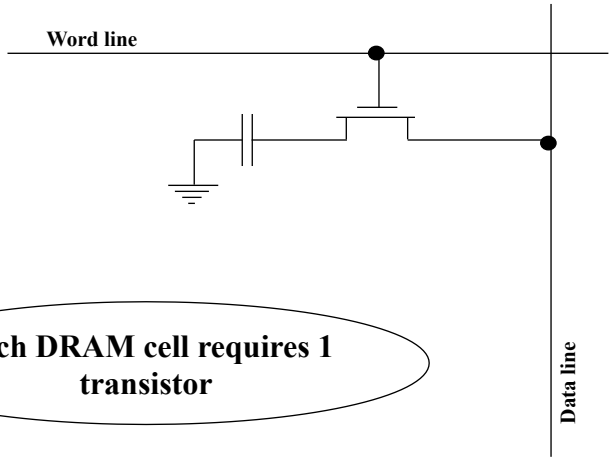
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Dynamic RAM cell



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Dynamic RAM cell



Each DRAM cell requires 1 transistor

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Behavior

Read operations:

- **by activating the word line, the capacitor is connected to the data line, which then assumes a value of 0 or 1 depending on the stored value.**

Write operations:

- **the word line activation causes the data line to connect to the capacitor, which is then charged or discharged, depending on the value of the line.**

A dedicated sensor connected to the data line can detect any voltage change caused by the charge / discharge of the capacitor, and produce the corresponding bit.

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Refresh

The charge contained in the capacitor tends to become 0 due to unavoidable dispersion currents.

Hence, to preserve the stored bit value, you need to periodically restore the charge (if any).

This can be done through a fictitious reading operations, in which the read value is not transmitted on the outside.

It is indispensable to keep the contents of each DRAM cell indefinitely.

Refreshing operations occupy a RAM for very short times (of a few percent).

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Refresh circuitry

In the past, it had to be run by some circuitry external to the DRAM module.

Currently it is always realized via dedicated internal circuitry.

It is therefore almost transparent to the user.

The refresh and normal access operations may interfere one with the other: in this case, the normal operation is temporarily suspended.

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Reliability

- **If a dynamic memory cell is affected by a charged particle, the stored charge may change, changing the stored value (*transient fault*).**
- **As the integration density increases, the size of the charge stored in each cell tends to decrease significantly, making memory more and more sensitive to faults (and thus less reliable).**
- **Most dynamic memories are protected for this purpose, for example through the use of protection codes.**

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Protection codes

To increase the reliability of DRAM memories, each word can be associated with some code bits.

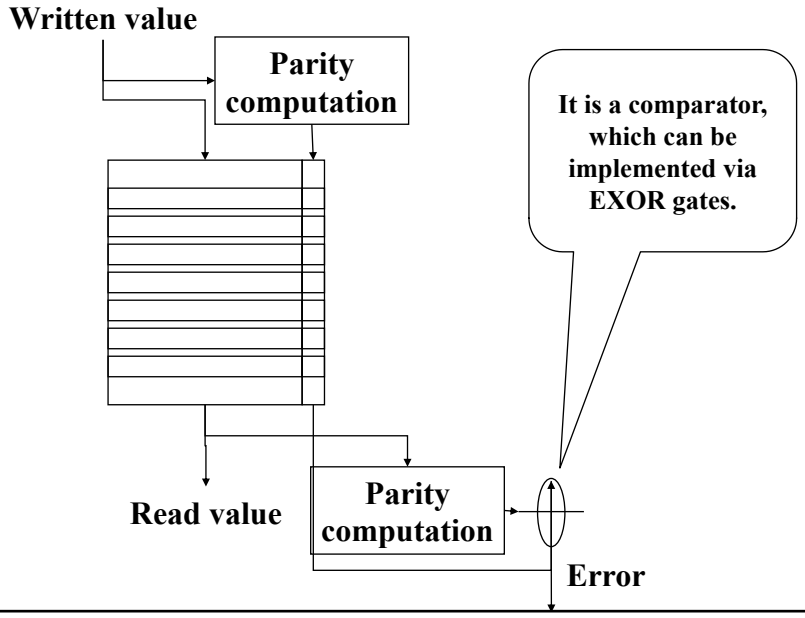
The simplest case of protection code is the parity code (1 bit).

Behavior:

- **when a write operation is performed, an ad hoc module computes the parity bit for the data to be written and stores it with the new value in a special column**
- **when reading the word, another module computes the parity code associated with the read value, and compares it with the stored one**
- **in case of diversity, an error signal is raised.**

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Parity code: architecture



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Hamming codes

Through Hamming codes it is possible not only to detect, but also to correct any error that occurs in a memory.

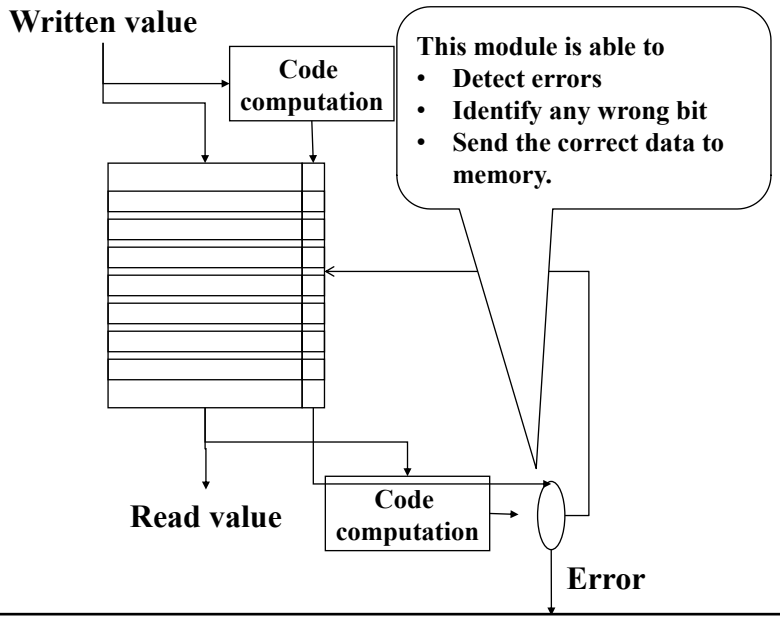
If *n* is the memory width, a Hamming code

- **Requires $1 + log_2 n$ code bits**
- **Can detect and correct all single errors**
- **Can detect (but not correct) all double errors.**

These codes are denoted as SECDED (*Single Error Correction Double Error Detection*).

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Hamming code: architecture



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Example

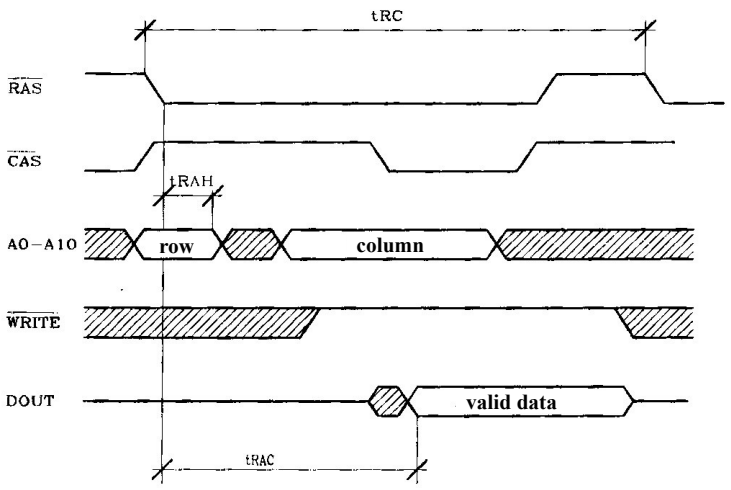
Let consider the AS4C1M16E5 DRAM produced by Alliance Semiconductor.

Characteristics:

- **1 M × 16 bit**
- **Square matrix organization**
- **Pins:**
  - **A_0 – A_9: address**
  - **RAS, CAS (when they are 0, they are active)**
  - **DQ1 – DQ16: data**
  - **WE: enables writing (active when 0)**
  - **OE: enables the outputs (active when 0).**

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Timing



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Read operations

- The row address is written on $A_0 – A_9$
- RAS is activated
- Wait for a time equal to t_{RAH}
- The column address is written on $A_0 – A_9$
- CAS is activated
- After a time equal to t_{RAC} the data are available on DOUT.

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Notes

- You can not start a new read or write cycle unless a tRC time (equal to 100 ns) has passed.
- Every 16 ms a complete refresh need to be performed, corresponding to the execution of 1,024 read operations with increasing line address.

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SRAMs vs. DRAMs

Static RAMs are (compared to dynamic RAMs):

- **faster (access times of 10 ns vs. 100 ns)**
- **more expensive (in terms of required silicon area, so less dense)**
- **easier to use**
- **more reliable.**

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